MILITARY SPECIFICATION

INDICATOR, COMPASS REPEATER, ROTATING DIAL, TYPE MW-2

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification covers certain design requirements and all performance requirements for procurement of one type of 26 volt, single phase, 400 cycle, rotating dial compass repeater indicator designated Type MW-2.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publication of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Method of
MIL-C-5015	Connectors, Electric, "AN" Type
MIL-D-5028	Drawings and Data Lists: Preparation of Manufacturers' (For Production Aeronautical and Associated Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-I-6181	Interference Limits and Tests; Aircraft Electrical and Electronic Equipment
MIL-S-6872	Soldering Process, General Specification for
MIL-L-6880	Lubrication of Aircraft, General Specification for
MIL-P-6906	Plates, Information and Identification
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-P-7936	Preparation for Delivery of Naval Aeronautical Equipment; General Specification for
MIL-A-8625	Anodic-Coatings, for Aluminum and Aluminum Alloys
MIL-C-17858	Compass System: Magnetic Slaved, Directional Gyro Stabilized, 4°/HR, Type MA-1

STANDARDS

Federal

FED. STD. No. 1	Standard for Laboratory Atmospheric Conditions for Testing
FED. STD. No. 595	Colors

FSC 6605

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Military

MIL-STD-130	Identification Marking of U. S. Military Property
MS28094	Indicator, Compass Repeater, Rotating Dial, Type MW-2
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33586	Metals, Definition of Dissimilar

Air Force-Navy Aeronautical

AN5727	Glass - Aircraft Instrument Cover

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards, Use of

(When requesting any of the applicable documents refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, 700 Robbins Avenue, Philadelphia 11, Pennsylvania, Attention: Code CDS)

3. REQUIREMENTS

3.1 Qualification. - The indicator furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein.

3.2 Selection of Government documents. - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with ANA Bulletin No. 143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts. - Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.2.2 Standard parts. - With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials. - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Critical materials. - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Metals. - Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to salt spray or atmospheric conditions to which the indicator may be subjected when in storage or during normal service life.

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3.3.2.1 Dissimilar metals. - Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in Standard MS33586 shall not be used in intimate contact with each other.

3.3.3 Nonmagnetic materials. - Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.4 Fungus-proof materials. - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.5 Corrosive fumes. - The materials used in the indicator shall not liberate deleterious fumes.

3.4 Design and construction. - The indicator shall be designed to repeat the aircraft azimuth heading as received from the output synchro of the MA-1 compass system conforming to Specification MIL-C-17858. The repeater synchro shall contain three stator coils fixed to the case and a rotor coil which shall be free to rotate. The dial of the instrument shall rotate freely thru 360 degrees and shall be connected to the shaft of the synchro rotor. The design shall be reasonably simple, to permit service maintenance, repair or overhaul without requiring special tools and fixtures. The indicator shall be constructed to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service.

3.4.1 Maintenance. - The design of the indicator shall be such as to accommodate to the greatest possible extent, disassembly, reassembly, and service maintenance by those tools and items of maintenance equipment which are normally available as commercial standards.

3.5 Performance. - The indicator shall perform satisfactorily when subjected to the tests specified in Section 4.

3.6 Case. - The outline dimensions of the case shall conform to Standard MS 28094. The design shall be such that the mechanism may be removed from the case and replaced without the use of special tools and fixtures.

3.6.1 Body and mounting flange. - The body and mounting flange of the case shall be made of lightweight nonmagnetic metal, uniform in texture, and shall have a smooth surface. The body shall be finished in a durable black finish and the mounting flange shall be finished with a durable lusterless black finish, Color No. 37038 of Federal Standard No. 595.

3.6.2 Reinforcement. - The inside of the case around each connection shall be of sufficient strength to prevent damage to the case when the connections are tightened during installation of the indicator.

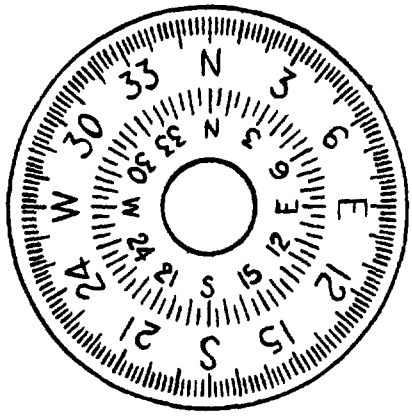
3.6.3 Cover glass. - The cover glass shall be in accordance with Standard AN5727 and shall be properly sealed, provision being made to prevent fogging or accumulation of moisture on the surface under extremes of atmospheric conditions. The distance between the cover glass and the mask shall be the minimum practicable and shall not exceed 0.125 inch.

3.6.4 Dial. - The indicator dial shall conform to figure 1, shall be of aluminum or aluminum alloy, and shall be securely fastened to the frame of the mechanism by not less than two screws in such a manner that it will not loosen or turn when the instrument is vibrated.

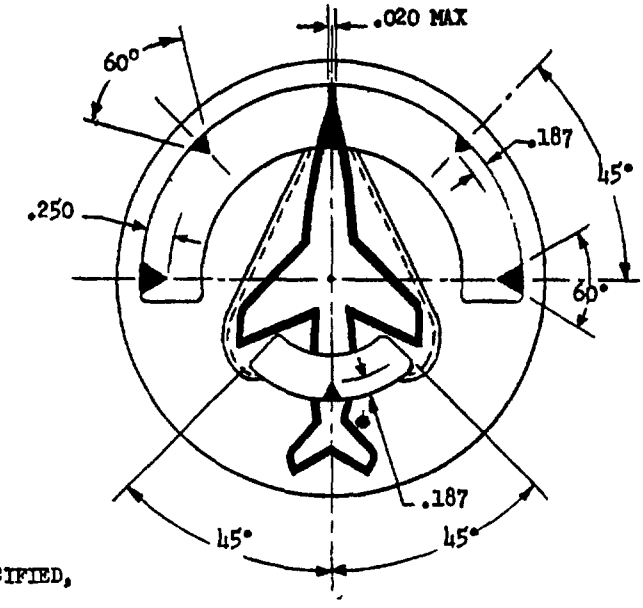
3.7 Screw threads. - Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification MIL-S-7742.

3.8 Wiring. - The electrical wiring system shall be insulated from the indicator case and shall conform to figure 2.

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DIAL



MASK

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ± .015, ANGLES ± 1°. NUMERALS SHALL DISTINCTLY INDICATE THE GRADUATION TO WHICH EACH APPLIES. LUBBER LINE AND POINTERS TO BE SPACED ACCURATELY WITHIN ±0° 15'.

MARKING	HEIGHT OR LENGTH ±.010	WIDTH OF LINE OR GRADUATION ±.005	FINISH
MAIN DIAL			LUSTERLESS WHITE
ALL NUMBERS AND LETTERS	.187		
30° GRADUATIONS	.187	.020	
10° GRADUATIONS	.187	.015	
2° GRADUATIONS	.125	.015	
RECIPROCAL DIAL			
ALL NUMBERS AND LETTERS	.125		
10° GRADUATIONS	.187	.020	
5° GRADUATIONS	.125	.015	
MASK			
AIRPLANE OUTLINE	----	.020	
LUBBER LINE AND AIRPLANE NOSE	----	----	
POINTERS	----	----	
BACKGROUND OF DIAL AND MASK	----	----	LUSTERLESS BLACK

Figure 1. Dial Markings

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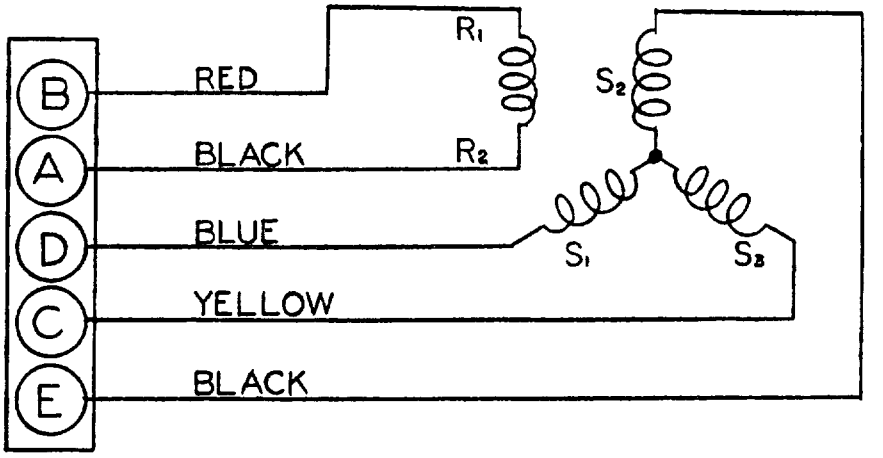


Figure 2. Wiring Diagram

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3.9 Electric receptacle. - The indicator shall be provided with an electric receptacle conforming to Specification MIL-C-5015, and shall be of the type specified in Standard MS28094. A hermetic type lead-in shall be required between the electric receptacle and the synchro within the sealed portion of the case.

3.10 Electrical characteristics. - The indicator shall be designed to receive its rotor excitation from a source of 26 volts, 400 cycles per second, single phase alternating current. The synchro stator output, measured from leg to leg, shall reach 11.8 volts as its maximum value.

3.11 Interchangeability. - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.12 Physical requirements. -

3.12.1 Weight. - The weight of the indicator shall not exceed 1.65 pounds, not including mounting nuts and screws.

3.12.2 Finish. - Protective coatings and finishes which will crack, chip or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.12.2.1 Aluminum alloy parts. - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

3.12.2.2 Iron and steel parts. - Where practicable, iron and steel parts shall be cadmium, chromium or nickel plated in accordance with Specifications QQ-P-416, QQ-C-320, and QQ-N-290, respectively.

3.12.3 Soldering. - Soldering shall be performed in accordance with Specification MIL-S-6872.

3.12.4 Lubrication. - Lubrication shall be accomplished in accordance with Specification MIL-L-6880, except that the selection of the lubricants used shall be approved by the procuring activity.

3.13 Marking. - All markings shall be durable to withstand the usage encountered in service. The form of the numerals and letters placed on the dial shall conform to Standard MS33558. Numerals and letters shall distinctly indicate the graduation to which each applies.

3.13.1 Dial and mask. - The dial and mask shall be marked as shown in figure 1. Lusterless white markings shall conform to Color No. 37875 of Federal Standard No. 595. Background of the dial and mask shall be lusterless black Color No. 37038 of Federal Standard No. 595.

3.13.1.1 Visibility of dial. - The dial numerals, at least 1/16 inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the instrument case.

3.13.2 Identification of product. - A nameplate conforming to Specification MIL-P-6906 shall be securely attached to the indicator case, and shall be marked in accordance with Standard MIL-STD-130.

3.13.3 Receptacle. - The electric receptacle code marking as specified in Standard MS28094 shall be legibly and permanently applied to the indicator case.

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3.14 Installation.

3.14.1 Installation instructions. - The contractor shall furnish with each indicator, one printed copy of instructions, with illustrations and diagrams, if necessary, covering the installation of the indicator. Previous to printing, two copies shall be furnished to the Bureau of Aeronautics for approval. The instructions shall be printed on 8-1/2 by 11 or 11 by 17 inch paper.

3.14.2 Mounting screws. - The contractor shall furnish sufficient mounting screws for installing the indicator. The screws shall be No. 6-32 NC-2 fillister head, non-magnetic, machine screws having a durable and dull black oxidized, or dull black nickel finish. The length shall be sufficient for mounting on a panel up to 3/8-inch thick.

3.14.3 Envelope. - An envelope furnished by the contractor containing the installation instructions and mounting screws shall be packaged with each indicator and shall be marked with the following information:

"IMPORTANT"

"THIS ENVELOPE CONTAINS INSTRUCTIONS AND MOUNTING SCREWS"

3.15 Workmanship. - The indicator, including all parts and accessories, shall be constructed and finished to produce an instrument free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests. - The inspection and testing of the indicators shall be classified as follows:

(a) Qualification tests. - Qualification tests are those tests performed on indicators submitted for qualification as a satisfactory product.

(b) Acceptance tests. - Acceptance tests are those tests performed on indicators manufactured and submitted for acceptance under contract.

4.2 Qualification tests. - The qualification tests of the indicators shall consist of all tests of this specification. The tests shall be conducted in the following sequence:

Individual tests
Sampling tests
Radio noise interference
Altitude
Low temperature exposure
High temperature exposure
Vibration failure
Humidity
Fungus resistance
Salt spray
Mounting lugs
Life

4.2.1 Qualification test samples. - Qualification test samples shall consist of three indicators of each manufacturer's part number submitted as samples for qualification testing. After obtaining authorization for submittal from the activity responsible for qualification, the samples shall be forwarded to the testing laboratory designated in letter of authorization. Each sample shall be plainly

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identified by a securely attached durable tag marked with the following information:

Sample for qualification test
Submitted by (name) (date) for qualification tests in
accordance with the requirements of Specification MIL-I-21685
under authorization (reference letter authorizing test)
Indicator, Compass Repeater, Rotating Dial, Type MW-2
Manufacturer's Part No.
Name of Manufacturer

4.3 Acceptance tests. - The acceptance tests of the indicator shall consist of the individual tests and the sampling tests of this specification.

4.3.1 Individual tests. - The individual tests of the indicator shall consist of the following tests conducted on each indicator:

Inspection
Dielectric strength
Scale error
Friction error
Synchro null

4.3.2 Sampling tests. - The sampling tests of the indicator shall consist of the following tests conducted on each sampling test sample:

Magnetic effect
Vibration error
Damping
Low temperature operation
High temperature operation
Phase rotation

4.3.2.1 Sampling test instructions. - Samples shall consist of three indicators selected at random by the Inspector from each lot of 100 or less indicators which have passed the Individual tests. A lot shall consist of identical indicators with the same manufacturer's part number, manufactured under substantially the same conditions and submitted at substantially the same time. Indicators which have been submitted to the sampling tests shall not be delivered on contract until they have been rebuilt and resubmitted to all Individual tests.

4.4 Rejection and retest. - Indicators which have been rejected as a result of the Individual tests may be reworked to correct the defects, or replaced, and resubmitted for acceptance. Failure of samples representing lots shall be cause for rejection of the lot represented. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units or lots rejected after retest shall not be resubmitted without specific approval of the procuring activity.

4.5 Test conditions.

4.5.1 Atmospheric conditions. - Unless otherwise specified, all tests required by this specification shall be made in accordance with the requirements of Federal Standard No. 1.

4.5.2 Level position. - Unless otherwise specified, the indicator shall be tested in its level position in which the upper mounting holes are on a horizontal line and the face is vertical.

4.5.3 Electrical power. - Unless otherwise specified, tests requiring the operation of the indicator shall be conducted utilizing a 26 ±2 volts, single phase, 400 ±5 cycles per second, alternating current. The indicator operating under these conditions shall be deemed properly connected.

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4.5.4 Standard test units. - The standard signal shall consist of a data transmitter, similar to that used for the output synchro of the MA-1 compass system conforming to Specification MIL-C-17858. The data transmitter shall be mounted so as to indicate angular displacement between rotor and stator from 0 to 360 degrees in steps of 0.1 degree. The standard data transmitter shall have minimum and known errors.

4.5.5 Tapping. - Unless otherwise specified, the indicator may be tapped or vibrated before a test reading is taken.

4.5.6 Vibration stand. - Whenever a vibration stand is specified it shall consist of a device which will vibrate at any desired frequency between 5 and 50 cycles per second and shall subject the indicator to such vibration that a point on the vibration stand shall describe, in a plane inclined 45 degrees to the horizontal plane, a circle of the diameter specified herein.

4.6 Test methods.

4.6.1 Inspection. - Each indicator shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

4.6.2 Dielectric strength. - A potential of 500 volts (root mean square) alternating current at commercial frequency shall be applied between isolated pins, and between pins and the case for a period of 5 seconds. There shall be no breakdown of insulation or any other permanent damage to the instrument as a result of this test.

4.6.3 Scale error. - The indicator shall be mounted with the case in its normal position and connected to the standard data transmitter. The indicator pointer shall be set to zero by rotating the rotor of the standard transmitter. The deviation between the standard data transmitter and the indicator shall be recorded on each 30 degree heading of the indicator through 360 degrees. The deviations shall be added algebraically and divided by the number of headings. The result shall be the index error and shall not exceed 1/2 degree. Any one reading of the indicator shall not differ from the data transmitter reading plus the index error by more than one degree. This test shall be conducted with the instrument subjected to a vibration in the vertical plane of 0.003 to 0.005 amplitude at a frequency of approximately 1750 cycles per minute.

4.6.4 Friction error. - The Scale error test shall be repeated with no vibration applied. The difference between the readings obtained in this test and the readings obtained in the Scale error test shall not exceed 1-1/2 degrees.

4.6.5 Synchro null. - Make electrical connections as shown in figure 3. With switches open, turn the rotor shaft until a minimum voltage is obtained on voltmeter V_1. The rotor will now be in the true electrical zero position or 180 degrees away. With the switches closed, voltmeter V_2 is then used to distinguish between the 15 volts which occur in the true electrical zero and the 38 volts which occur in the false electrical zero. To set electrical zero accurately, a sensitive vacuum tube voltmeter must then be inserted between the yellow and blue leads, (see figure 2); the switches opened, and a fine setting made on the rotor position to obtain the minimum voltage, approximately 10 millivolts.

4.6.6 Magnetic effect. - The indicator shall be revolved about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from the bar magnet, and in a vertical plane which is a perpendicular bisector of the North-South axis of the bar magnet. Starting directly under the compass, the indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees from the initial position. At each of these positions the indicator shall be rotated on its horizontal axis until it is in its normal operating position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed one degree. This test shall be performed first with no voltage applied to the indicator and then shall be repeated with the indicator operating.

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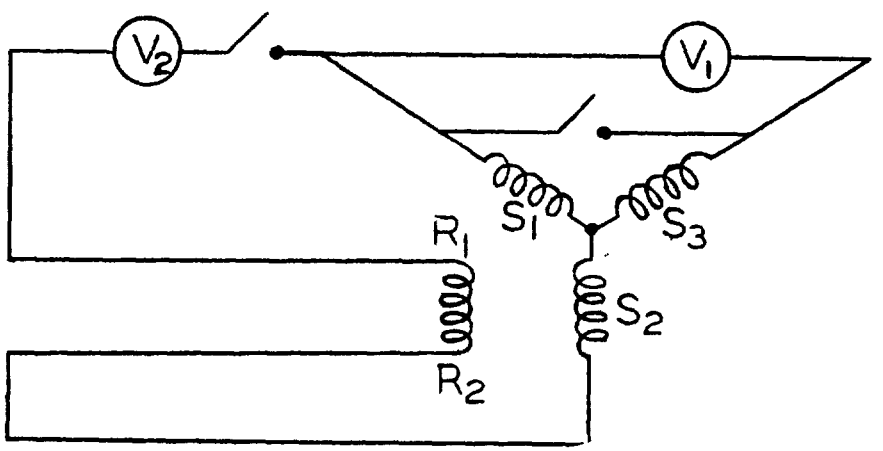


Figure 3. Synchro Null Test Wiring

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4.6.7 Vibration error.-- The indicator shall be mounted on the vibration stand in its normal operating position. The indicator shall be operated and shall be subject to vibration with a circular motion in a plane inclined 45 degrees to the horizontal plane with a circle diameter of between 0.009 and 0.011 of an inch. The frequency of applied variation shall be varied slowly from 5 to 50 cycles per second. No condition of applied variation shall cause vibration errors involving pointer oscillation or variation in excess of one degree.

4.6.8 Damping. - Two indicators shall be connected to the standard data transmitter, with their pointers displaced (within the limits of 45 to 90 degrees) in opposite directions from the locked standard data transmitter setting, and instantly energized from the alternating current buss at 26 volts, 400 cycles per second. The time limit for pointer oscillation to damp out, if present, shall not exceed 4 seconds.

4.6.9 Low temperature operation. (This test may be combined with the Low temperature exposure test at the discretion of the testing agency). - The indicator shall be properly connected except that no power shall be applied. The indicator shall then be subjected to an ambient temperature of -54° C (-65° F) for a period of 4 hours. During the last 15 minutes of the 4-hour period, and with the indicator connected to the standard data transmitter, power shall be applied to the indicator. At the end of the 4-hour period and with the temperature maintained at -54° C (-65° F) the transmitter shall be rotated slowly through 360 degrees. The indicator pointer shall operate around the 360 degree dial without sticking or lagging by more than 3 degrees.

4.6.10 High temperature operation. (This test may be combined with the High temperature exposure test at the discretion of the testing agency). - The indicator shall be properly connected except that no power shall be applied. The indicator shall then be subjected to an ambient temperature of 71° C (160° F) for a period of 4 hours. During the last 15 minutes of the 4-hour period, power shall be applied to the indicator. At the end of the 4-hour period and with the temperature maintained at 71° C (160° F) the indicator shall be subjected to and meet the requirements specified in the Individual tests. The indicator shall then be allowed to cool to room temperature and shall be subjected to and meet the requirements of the Scale error and Friction error tests.

4.6.11 Phase rotation. - With the indicator dial set on the North heading, and pins "A" and "E" connected together, the voltage between pins "B" and "D" shall decrease before increasing as the heading is increased.

4.6.12 Radio noise interference. - The electrical system of the indicator shall be tested for conducted and radiated radio noise interference in accordance with Specification MIL-I-6181.

4.6.13 Altitude. - The indicator properly connected with no power applied shall be tested in accordance with Procedure II of Specification MIL-E-5272 except that the chamber shall be reduced to 0.810 inch of mercury (80,000 feet) for a period of one hour. During the last 15 minutes of the one hour period, power shall be applied to the indicator. At the end of the one hour period and with the temperature and pressure maintained as specified, the indicator shall be subjected to and meet the requirements specified in the Low temperature operation test.

4.6.14 Low temperature exposure. - The indicator shall be tested in accordance with Procedure II of Specification MIL-E-5272. The Individual tests shall be repeated at room temperature.

4.6.15 High temperature exposure. - The indicator shall be tested in accordance with Procedure I of Specification MIL-E-5272. The Individual tests shall be repeated at room temperature.

4.6.16 Vibration failure. --The indicator shall be tested in accordance with Procedure V of Specification MIL-E-5272. The indicator shall be subjected to and meet the requirements specified in the Individual tests.

4.6.17 Humidity. - The indicator shall be tested in accordance with Procedure I of Specification MIL-E-5272. The indicator shall be subjected to and meet the requirements specified in the Individual tests.

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4.6.18 Fungus resistance. - The indicator shall be tested in accordance with Procedure I of Specification MIL-E-5272, except that the test period shall be 14 days. At the option of the qualifying agency, tests of the component parts of the indicator may be accepted in lieu of, or in addition to, tests of the assembled instrument.

4.6.19 Salt spray. - The indicator shall be tested in accordance with Procedure I of Specification MIL-E-5272. The indicator shall be subjected to and meet the requirements specified in the Individual tests.

4.6.20 Mounting lugs. - The indicator case shall be mounted face downward on the removable head of a suitable testing machine with the face of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A load of 175 pounds shall be applied for one minute to each lug in a direction toward the front of the case. The lugs shall withstand the applied load without fracture and there shall be no damage to any part of the indicator.

4.6.21 Life. - The indicator shall be subjected to a life test of 1000 hours. These 1000 hours shall include approximately 100 hours accumulated on the indicator during other tests. The instrument shall be bench mounted and operated normally during this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. - The indicator shall be packaged in accordance with Specification MIL-P-7936. The method of preservation shall be in accordance with Specification MIL-P-116, Method IId using metal reusable containers.

5.1.2 Level C. - When this level is required, packaging shall be in accordance with standard commercial practice.

5.2 Packing.

5.2.1 Levels A and B. - The indicator shall be packed in accordance with Specification MIL-P-7936. The level or levels required shall be as specified in the invitation for bid or contract.

5.2.2 Level C. - The indicator shall be packed to insure that the shipment arrives in a satisfactory condition at destination. The shipment shall conform to the applicable carriers rules and regulations in effect at the time of shipment.

5.3 Marking. - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

6. NOTES.

6.1 Intended use. - The compass repeater indicator covered by this specification is intended for use in conjunction with the MA-1 compass system and other directional gyro roll stabilized compass systems.

6.2 Ordering data. - Requisitions, contracts and orders shall specify the MS part number of the indicator desired, the quantity, and whether overseas packing is to be furnished.

6.3 Qualification. - With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.

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6.3.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Aeronautics, Department of the Navy, Washington 25, D. C.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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